

Retail Sales & Customer Analytics using SQL + Power BI

Project Overview

This project analyzes retail sales data using SQL and Power BI to uncover business insights related to customer behavior, product performance, sales trends, profitability, and operational KPIs.

Dataset Used:

- Sample - Superstore.csv

Tech Stack:

- SQL (MySQL/PostgreSQL)
- Power BI
- Python (optional preprocessing)
- GitHub

Recommended GitHub Repository Name

sql-data-analysis-projects

Recommended Repository Structure

```
sql-data-analysis-projects/  
|  
|— datasets/  
|   |— Sample - Superstore.csv  
|  
|— schema/  
|   |— create_tables.sql  
|  
|— queries/  
|   |— basic_queries.sql  
|   |— joins_queries.sql  
|   |— cte_queries.sql  
|   |— window_functions.sql  
|   |— case_statements.sql  
|   |— business_problems.sql  
|
```

```
├── dashboard/
│   └── retail_dashboard.pbix
├── screenshots/
│   ├── dashboard_overview.png
│   ├── customer_analysis.png
│   └── sales_trends.png
├── insights/
│   └── business_insights.md
└── README.md
```

Dataset Columns Used

Main columns from Superstore dataset:

- Order ID
- Order Date
- Ship Date
- Ship Mode
- Customer ID
- Customer Name
- Segment
- Country
- City
- State
- Region
- Product ID
- Category
- Sub-Category
- Product Name
- Sales
- Quantity
- Discount
- Profit

SQL DATABASE SETUP

Step 1: Create Database

```
CREATE DATABASE retail_sales_analysis;
USE retail_sales_analysis;
```

CREATE TABLE QUERY

create_tables.sql

```
CREATE TABLE superstore (  
    row_id INT,  
    order_id VARCHAR(50),  
    order_date DATE,  
    ship_date DATE,  
    ship_mode VARCHAR(50),  
    customer_id VARCHAR(50),  
    customer_name VARCHAR(100),  
    segment VARCHAR(50),  
    country VARCHAR(50),  
    city VARCHAR(100),  
    state VARCHAR(100),  
    postal_code VARCHAR(20),  
    region VARCHAR(50),  
    product_id VARCHAR(50),  
    category VARCHAR(100),  
    sub_category VARCHAR(100),  
    product_name VARCHAR(255),  
    sales DECIMAL(10,2),  
    quantity INT,  
    discount DECIMAL(5,2),  
    profit DECIMAL(10,2)  
);
```

DATA IMPORT

MySQL Import

```
LOAD DATA INFILE 'Sample - Superstore.csv'  
INTO TABLE superstore  
FIELDS TERMINATED BY ','  
ENCLOSED BY ''''  
LINES TERMINATED BY '\n'  
IGNORE 1 ROWS;
```

BASIC SQL QUERIES

basic_queries.sql

Total Sales

```
SELECT ROUND(SUM(sales),2) AS total_sales  
FROM superstore;
```

Total Profit

```
SELECT ROUND(SUM(profit),2) AS total_profit  
FROM superstore;
```

Total Orders

```
SELECT COUNT(DISTINCT order_id) AS total_orders  
FROM superstore;
```

Average Order Value

```
SELECT ROUND(SUM(sales)/COUNT(DISTINCT order_id),2) AS avg_order_value  
FROM superstore;
```

JOINS QUERIES

joins_queries.sql

Top Customers by Sales

```
SELECT customer_name,  
       ROUND(SUM(sales),2) AS total_sales  
FROM superstore  
GROUP BY customer_name  
ORDER BY total_sales DESC  
LIMIT 10;
```

Most Profitable Products

```
SELECT product_name,  
       ROUND(SUM(profit),2) AS total_profit
```

```
FROM superstore
GROUP BY product_name
ORDER BY total_profit DESC
LIMIT 10;
```

Region-wise Sales

```
SELECT region,
       ROUND(SUM(sales),2) AS total_sales
FROM superstore
GROUP BY region
ORDER BY total_sales DESC;
```

CTE QUERIES

cte_queries.sql

Monthly Sales Trend

```
WITH monthly_sales AS (
    SELECT MONTH(order_date) AS month_number,
           ROUND(SUM(sales),2) AS total_sales
    FROM superstore
    GROUP BY MONTH(order_date)
)
SELECT *
FROM monthly_sales
ORDER BY total_sales DESC;
```

Category Performance

```
WITH category_sales AS (
    SELECT category,
           ROUND(SUM(sales),2) AS total_sales,
           ROUND(SUM(profit),2) AS total_profit
    FROM superstore
    GROUP BY category
)
SELECT *
FROM category_sales
ORDER BY total_sales DESC;
```

WINDOW FUNCTIONS

window_functions.sql

Sales Ranking

```
SELECT category,
       product_name,
       sales,
       RANK() OVER(PARTITION BY category ORDER BY sales DESC) AS sales_rank
FROM superstore;
```

Running Total Sales

```
SELECT order_date,
       sales,
       SUM(sales) OVER(ORDER BY order_date) AS running_total_sales
FROM superstore;
```

Previous Month Comparison

```
SELECT MONTH(order_date) AS month_number,
       ROUND(SUM(sales),2) AS monthly_sales,
       LAG(ROUND(SUM(sales),2)) OVER(ORDER BY MONTH(order_date)) AS
previous_month_sales
FROM superstore
GROUP BY MONTH(order_date);
```

CASE STATEMENTS

case_statements.sql

Customer Segmentation

```
SELECT customer_name,
       ROUND(SUM(sales),2) AS total_sales,
       CASE
           WHEN SUM(sales) > 10000 THEN 'High Value Customer'
           WHEN SUM(sales) > 5000 THEN 'Medium Value Customer'
           ELSE 'Low Value Customer'
       END AS customer_segment
FROM superstore
```

```
GROUP BY customer_name
ORDER BY total_sales DESC;
```

Profitability Status

```
SELECT product_name,
       ROUND(SUM(profit),2) AS total_profit,
       CASE
         WHEN SUM(profit) > 0 THEN 'Profitable'
         ELSE 'Loss Making'
       END AS profit_status
FROM superstore
GROUP BY product_name;
```

BUSINESS PROBLEMS

business_problems.sql

Top 10 Products by Revenue

```
SELECT product_name,
       ROUND(SUM(sales),2) AS total_revenue
FROM superstore
GROUP BY product_name
ORDER BY total_revenue DESC
LIMIT 10;
```

Top 10 Loss Making Products

```
SELECT product_name,
       ROUND(SUM(profit),2) AS total_loss
FROM superstore
GROUP BY product_name
HAVING total_loss < 0
ORDER BY total_loss ASC
LIMIT 10;
```

Most Profitable Category

```
SELECT category,
       ROUND(SUM(profit),2) AS total_profit
FROM superstore
```

```
GROUP BY category
ORDER BY total_profit DESC;
```

Repeat Customers

```
SELECT customer_name,
       COUNT(DISTINCT order_id) AS total_orders
FROM superstore
GROUP BY customer_name
HAVING COUNT(DISTINCT order_id) > 5
ORDER BY total_orders DESC;
```

Regional Performance

```
SELECT region,
       ROUND(SUM(sales),2) AS total_sales,
       ROUND(SUM(profit),2) AS total_profit
FROM superstore
GROUP BY region
ORDER BY total_profit DESC;
```

Highest Sales Month

```
SELECT MONTHNAME(order_date) AS month_name,
       ROUND(SUM(sales),2) AS total_sales
FROM superstore
GROUP BY MONTHNAME(order_date)
ORDER BY total_sales DESC;
```

POWER BI DASHBOARD

Dashboard Pages

1. Sales Overview

KPIs:

- Total Sales
- Total Profit
- Total Orders
- Average Order Value

Charts:

- Monthly Sales Trend

- Region-wise Sales
 - Category-wise Revenue
-

2. Customer Analysis

Charts:

- Top Customers
 - Customer Segmentation
 - Repeat Customers
 - Segment-wise Revenue
-

3. Product Analysis

Charts:

- Top Products
 - Loss Making Products
 - Category Profitability
 - Sub-category Analysis
-

4. Regional Analysis

Charts:

- State-wise Sales
 - Region-wise Profit
 - City-wise Performance
-

BUSINESS INSIGHTS

business_insights.md

```
# Business Insights
```

```
## Key Findings
```

- Technology category generated the highest revenue.
 - West region showed the highest overall sales.
 - Some furniture products generated losses despite high sales.
 - A small percentage of customers contributed majority revenue.
 - Repeat customers significantly increased profitability.
-

README.md

Retail Sales & Customer Analytics using SQL + Power BI

Project Overview

This project analyzes retail sales data to uncover business insights related to customer behavior, sales performance, profitability, and operational KPIs using SQL and Power BI.

Dataset

- Superstore Sales Dataset

Tools Used

- SQL
- Power BI
- Excel

SQL Concepts Used

- Joins
- CTEs
- Window Functions
- Case Statements
- Aggregate Functions

Business Problems Solved

- Top-selling products
- Customer segmentation
- Regional performance analysis
- Profitability analysis
- Repeat customer analysis

Dashboard Features

- Sales Dashboard
- Customer Insights Dashboard
- Product Analysis Dashboard
- Regional Performance Dashboard

Key Insights

- Technology category generated highest sales.
- West region generated maximum profit.
- Certain furniture products caused losses.
- Repeat customers contributed major revenue.

Author

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WHAT TO ADD IN RESUME

Retail Sales & Customer Analytics using SQL + Power BI

- Developed a complete retail sales analytics solution using SQL and Power BI to analyze customer behavior, sales trends, profitability, and operational KPIs.
- Implemented advanced SQL concepts including joins, CTEs, window functions, and case statements to solve real-world business problems.
- Built interactive Power BI dashboards for sales monitoring, customer segmentation, and regional performance analysis.

WHAT TO ADD IN LINKEDIN PROJECTS

Developed an end-to-end Retail Sales & Customer Analytics project using SQL and Power BI. Performed sales trend analysis, customer segmentation, profitability analysis, and KPI reporting using advanced SQL concepts such as joins, CTEs, window functions, and case statements.

GITHUB PINNED REPOSITORIES RECOMMENDED

1. [sql-data-analysis-projects](#)
2. [powerbi-business-dashboard](#)
3. [promptstream-qa-dashboard](#)
4. [EmotionSenseAI](#)
5. [employee_analysis](#)
6. [kea-ai-orchestration-system](#)

FINAL RECRUITER IMPACT

This project demonstrates:

- SQL expertise
- Business analytics skills
- Dashboarding
- KPI reporting
- Problem-solving
- Real-world data analysis
- Data storytelling
- Power BI integration

This is highly valuable for:

- Data Analyst roles
- BI Analyst roles
- Reporting Analyst roles
- SQL Analyst roles
- Operations Analyst roles